

FLARE LOOP RADIATIVE HYDRODYNAMICS. VI. CHROMOSPHERIC EVAPORATION DUE TO HEATING BY NONTHERMAL ELECTRONS

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ABSTRACT

The response of the solar chromosphere to flare heating by nonthermal electrons is examined. A number of interesting phenomena appear in our numerical solutions of the equations of hydrodynamics and radiative transfer. Here we discuss one aspect of these results: the phenomenon of chromospheric evaporation. We present results for a range of heating fluxes and show how these may be understood in simple terms.

Our major conclusions are as follows: (1) There is an energy flux threshold for "explosive" evaporation. Explosive evaporation occurs when the upper chromosphere is unable to radiate the flare energy deposited there, and is therefore heated rapidly to coronal temperatures. Energy fluxes less than this threshold produce "gentle" evaporation, in which the chromosphere is eaten away by conduction at a much slower rate. (2) The expansion velocity of explosively evaporated plasma cannot exceed $\sim 2.35c_s$, where c_s is the sound speed in the evaporated material. (3) We derive a simple analytic model for the temporal variation of velocity in explosively evaporated plasma. This "gasbag" model, based on isothermal expansion of an impulsively heated mass of plasma, is used successfully to reproduce our own numerical results, as well as those of MacNeice *et al.* (1984). (4) The lower transition region, in both gentle and explosive evaporation, quickly reaches a quasi-steady balance between conduction and radiation, so that the conductive flux at 10^5 K is given by $3.42 \times 10^5 P$ ergs $\text{cm}^{-2} \text{s}^{-1}$, where $P(\text{dyn cm}^{-2})$ is the pressure in the flare transition region. In the case of explosive evaporation, a short powerful pulse of EUV radiation is emitted from plasma with temperatures near 10^5 K during the adjustment to this equilibrium.

Subject headings: hydrodynamics — radiative transfer — Sun: chromosphere — Sun: flares

I. INTRODUCTION

We define the term *chromospheric evaporation* to mean the heating of chromospheric material sufficiently strongly that, as a result of thermal instability, it jumps to coronal temperatures; the heated material consequently expands upward into the corona. We specifically exclude upward motion of plasma due to nonhydrodynamic, e.g., magnetic forces. Chromospheric evaporation in the context of solar flares is frequently invoked to explain the increase in coronal emission measure during the thermal phase of flares.

The chromospheric evaporation interpretation is not universally accepted, however. Cheng, Feldman, and Doschek (1981) argue that pressure gradients observed in flares are not consistent with upward motion of soft X-ray emitting material. They argue instead that the increase in emission measure might be due to magnetic compression in the corona from increased currents along magnetic loop structures. However, recent observations by the *Solar Maximum Mission* (SMM) satellite do support the chromospheric evaporation picture. Acton *et al.* (1982) demonstrate that the appearance of soft X-ray emission measure is accompanied by a loss of chromospheric material that is more than adequate to account for the soft X-rays. Antonucci *et al.* (1982) and Antonucci and Dennis (1983) find blueshifted components of emission lines of Ca XIX and Fe XXV, which indicate the upward motion of soft X-ray emitting material, presumably representing chromospheric evaporation. We show elsewhere (Fisher, Canfield, and McClymont 1984) that the velocity amplitudes corresponding to these blueshifted components are consistent with chromospheric evaporation.

Acton *et al.* (1982), in their discussion of evidence for

chromospheric evaporation, distinguish between two different evaporation mechanisms. Gentle evaporation during the thermal phase of flares is an inevitable consequence of the large conductive flux that must be present in hot thermal flare loops. Antiochos and Sturrock (1978) developed a simple model of gentle evaporation based on a balance between conductive flux and enthalpy flux. Their model, which is valid for small velocities, has been generalized by Craig and McClymont (1984) to include the effect of radiation.

In contrast, Brown (1973) and Lin and Hudson (1976) consider "explosive" evaporation in flaring chromospheres heated impulsively by large fluxes of nonthermal electrons (the "thick-target" model). In this case, evaporation is very rapid and may form upward-moving shocks in the corona. Explosive evaporation occurs when a finite depth of chromospheric plasma is heated *en masse* to a high temperature. If the heating time scale is shorter than the hydrodynamic expansion time, the pressure rises greatly and the subsequent expansion is "explosive."

One of our goals in modelling the hydrodynamic and radiative response of solar flare loops to fluxes of nonthermal electrons is to study chromospheric evaporation. We are particularly interested in explosive evaporation during the impulsive phase. In this paper, we use the results of our numerical simulations described in Paper V (Fisher, Canfield, and McClymont 1985a) as well as other thick-target calculations to draw some general conclusions about evaporation in electron-heated flares. In § II, we discuss the heating threshold for explosive evaporation. An upper limit to evaporation velocities is derived in § III. We introduce a simple analytic model of explosive evaporation in § IV and discuss the structure of the flare transition region in § V. We end by summarizing the most important points in § VI.

II. THE "EXPLOSIVE" HEATING FLUX THRESHOLD

The threshold for explosive evaporation may be calculated approximately by considering the local balance between flare heating and radiation in the upper chromosphere.

If the energy deposition rate is just insufficient to raise the chromospheric temperature past the peak radiative loss rate near 10^5 K, the pressure can be raised by at most a factor of 10 ($P \propto T$ at constant density). The heated atmosphere then expands relatively slowly and cannot reach coronal temperatures until either its density—and therefore radiative loss rate—decreases sufficiently that its temperature rises above 10^5 K, or the moving transition region, driven by conductive evaporation, engulfs it.

A higher heating flux, however, raises the chromospheric temperature above 10^5 K: since the radiative loss rate falls at higher temperatures, the plasma can be quickly heated to coronal temperatures. In this case the rise in pressure is at least a factor of $\sim 10^2$, and perhaps $\sim 10^3$, for a temperature change from 10^4 K to 10^6 – 10^7 K. These cases are illustrated schematically in Figure 1.

In order to produce the large overpressure required for explosive evaporation, an additional requirement must be met: the heating time scale in the evaporating region should be less than the hydrodynamic expansion time scale,

$$3kT/Q \lesssim L_0/c_s, \quad (1)$$

where Q is the flare heating rate per particle, T is the final temperature of the heated plasma, c_s is the corresponding sound speed, and L_0 is the length of the heated high-pressure region.

We now derive the threshold heating flux for explosive evaporation. Although this particular derivation is tailored to the specific heating function adopted in Paper V, the approach

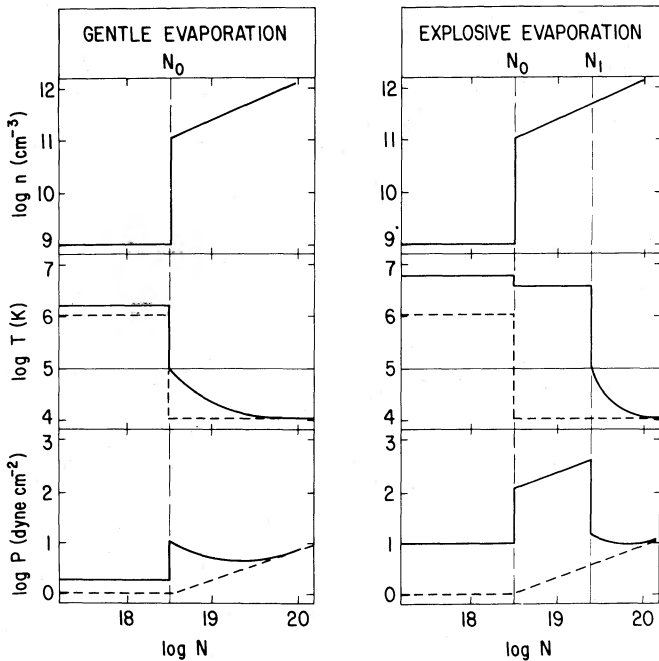


FIG. 1.—Diagram showing an idealized distribution of density, temperature, and pressure for both gentle and explosive evaporation. The time chosen is shortly after heating by energetic electrons has commenced, but before the material has moved. The dashed curves show the preflare state.

is quite general and applicable to a wide range of flare heating functions. Since the density—and therefore the radiative loss rate at a given temperature—increases with depth in the chromosphere, while the heating rate in general decreases, we define the threshold heating rate to be that which balances the radiative loss rate in the uppermost layer of the chromosphere, just below the preflare transition region. Thus the threshold is exceeded if $N_1 > N_0$, where N_0 is the column depth of the preflare transition region, and N_1 is the column depth of the flare transition region (at the base of the evaporated region), given by

$$Q(N_1) = n(N_1)f(T = 10^5 \text{ K}), \quad (2)$$

where $Q(N)$ is the flare heating rate per particle, $n(N)$ is the preflare density, and $f(T)$ is the optically thin radiative loss function, which we assume to peak at 10^5 K, with $f(T = 10^5 \text{ K}) \approx 7 \times 10^{22} \text{ ergs cm}^3 \text{ s}^{-1}$. To a good approximation we can treat each leg of the coronal loop as vertical, so that

$$n(N) = (P_c + mgN)/[(1+x)kT_{ch}], \quad (3)$$

where P_c , T_{ch} , and x are the preflare apex pressure and preflare chromospheric temperature and ionized fraction, respectively. We consider heating by Coulomb collisions of a flux of non-thermal electrons, injected near the loop apex, with a spectral index δ , low-energy cutoff E_c , and total electron energy flux F_e . For column depths $N > N_c$, where N_c is the penetration depth of electrons of energy E_c , the heating rate falls off as $(N/N_c)^{-\delta/2}$, but for $N \lesssim N_c$ the heating rate is approximately constant (see § III, Paper V). For the simulations discussed here, we find that $N_1 \lesssim N_c$; i.e., the flare transition zone forms in a region of near constant heating rate. We therefore introduce the constant N^* and write

$$Q(N) \approx F_e/N^*, \quad (4)$$

where for the parameters assumed ($E_c = 20 \text{ keV}$, $\delta = 4$), $N^* \approx 10^{20} \text{ cm}^{-2}$. Since $N_c \propto E_c^2$, the choice of a lower energy cutoff E_c might have resulted in $N_1 > N_c$; in this case the analysis can be carried out using the form of $Q(N)$ appropriate to $N > N_c$; see Lin and Hudson (1976) and McClymont, Canfield, and Fisher (1984).

Using equations (2), (3), and (4), we find the criterion for the threshold heating flux:

$$F_e > F_{\text{crit}} \equiv N^*f(10^5 \text{ K})(P_c + mgN_0)/[(1+x)kT_{ch}], \quad (5)$$

when N_0 is the column depth of the preflare transition region. If F_e exceeds F_{crit} , then N_1 , the column depth of the flare transition region, is given by

$$(N_1/N_0) = (F_e/F_{\text{crit}})[1 + P_c/(mgN_0)] - P_c/(mgN_0). \quad (6)$$

In the preflare loop atmosphere described in Paper V, the coronal pressure P_c is roughly 0.3 dyn cm^{-2} , and the column depth of the preflare transition region N_0 is about $3 \times 10^{18} \text{ cm}^{-2}$. Although a value of $T_{ch} \approx 10^4 \text{ K}$ should be used in equation (6) to calculate the column depth of the flare transition region for heating fluxes well above the threshold, the appropriate value of T_{ch} to use in defining the threshold flux is roughly $2 \times 10^4 \text{ K}$ (with $x = 1$), due to the presence of a Ly α plateau at the base of the transition region in the VAL/F model (Vernazza, Avrett, and Loeser 1981). Inserting these values in equation (5) yields $F_{\text{crit}} \approx 7 \times 10^9 \text{ ergs cm}^{-2} \text{ s}^{-1}$. The simulation with F_{20} (F_e with a 20 keV cutoff) = $10^9 \text{ ergs cm}^{-2} \text{ s}^{-1}$ ($F_e < F_{\text{crit}}$) does indeed show strikingly different behavior from

the runs $F_{20} = 3 \times 10^{10}$ ergs cm $^{-2}$ s $^{-1}$, $F_{20} = 5 \times 10^{10}$ ergs cm $^{-2}$ s $^{-1}$, and $F_{20} = 3 \times 10^{11}$ ergs cm $^{-2}$ s $^{-1}$, which all have $F_{20} > F_{\text{crit}}$ (see Figs. 2–6, Paper V).

In the case $F_{20} = 10^9$ ergs cm $^{-2}$ s $^{-1}$, the amount of evaporated material does not exceed 10^{17} cm $^{-2}$ during the simulation, which in this case was run to 25 s beyond the 5 s heating pulse. Clearly, this is a case of gentle evaporation.

The $F_{20} = 10^{10}$ ergs cm $^{-2}$ s $^{-1}$ case represents a value of F_{20} just above the threshold. In this case, the amount of material explosively evaporated is only about 6×10^{17} cm $^{-2}$ and represents a very thin layer at the top of the chromosphere. Since this is just a few times thicker than the transition region, its hydrodynamic response is not much different from that of the thin layer of chromospheric material evaporated by a moving conduction front; i.e., the condition (1), which is necessary for explosive evaporation, does not hold. This case, therefore, more resembles “gentle” evaporation than explosive evaporation, even though the heating rate exceeds the threshold value.

All three runs with high energy fluxes show a very rapid increase in the amount of evaporated material (Fig. 2), as well as a rapid pressure rise in the evaporated region (Figs. 2–6, Paper V). Explosive evaporation is accomplished in $\lesssim 1$ s, at which time 4.9×10^{18} cm $^{-2}$, 1.1×10^{19} cm $^{-2}$, and 2.1×10^{19} cm $^{-2}$ have been evaporated by the nonthermal electron fluxes of 3×10^{10} ergs cm $^{-2}$ s $^{-1}$, 5×10^{10} ergs cm $^{-2}$ s $^{-1}$, and 10^{11} ergs cm $^{-2}$ s $^{-1}$, respectively. These column masses are about half the amounts predicted by equation (6) with $T_{\text{ch}} = 10^4$ K and $x = 1$, but the discrepancy is reduced to about 20% if the true preflare temperature and ionized fraction evaluated at the flare transition region are inserted in equation (6).

Following explosive evaporation, there is a steady, but much slower, increase in the mass of evaporated plasma, due to conductively driven evaporation, just as in gentle evaporation. On the short time scales studied here, explosive evaporation is far

more important, both in terms of the amount of evaporated material produced and in terms of its hydrodynamic consequences, than is conductive evaporation. However, on the time scale of the flare as a whole, i.e., minutes rather than seconds, this is *not* true (Ricchiuzzi and Canfield 1983; Canfield and Gunkler 1985).

III. UPPER LIMIT TO EXPLOSIVE EVAPORATION VELOCITIES

If the upper chromosphere is heated rapidly to coronal temperatures, the resulting pressure gradients will tend to drive mass flows into both the corona and the residual chromosphere. In this section, we derive an upper limit to the outflow velocity of the evaporated material. This upper limit is not mechanism dependent and should be quite general, provided that the assumptions used in deriving it can be justified. To describe the upward expansion we use the Eulerian equation of motion

$$mn(\partial v/\partial t + v\partial v/\partial z) = -\partial P/\partial z, \quad (7)$$

where we have ignored gravity since we are concerned with expansion through a distance small compared with the coronal gravitational scale height. If the initial conditions and the flare heating rate are such that the temperature in the ambient corona and evaporating region increases monotonically with height, i.e.,

$$\partial T/\partial z \geq 0, \quad (8)$$

where z is measured downward, then it follows that

$$\frac{\partial P}{\partial z} = 2k \left[T \left(\frac{\partial n}{\partial z} \right) + n \left(\frac{\partial T}{\partial z} \right) \right] \leq 2kT \frac{\partial n}{\partial z} \leq 2kT_{\text{max}} \frac{\partial n}{\partial z}, \quad (9)$$

where T_{max} is the temperature at the top of the expanding region, and we assume $\partial n/\partial z \geq 0$, consistent with preflare hydrostatic equilibrium. We expect equation (9) to give a close upper bound to the pressure gradient, from the bottom of the expanding region up into the corona, since most of the pressure differential is due to the density difference between the expanding region and the ambient corona. Once the heated chromospheric plasma has reached coronal temperatures, thermal conduction helps maintain its temperature close to the temperature of the low-density corona above. Thus equation (7) may be written

$$v\partial v/\partial v + [2kT_{\text{max}}/(mn)]\partial n/\partial z \geq -\partial v/\partial t,$$

or

$$(\partial/\partial z)[(\frac{3}{5})c_s^2 \ln(n) + (\frac{1}{2})v^2] \geq -\partial v/\partial t, \quad (10)$$

where $c_s = [(10/3)kT_{\text{max}}/m]^{1/2}$ is an upper limit to the sound speed in the corona. Integrating from some point in the preflare corona to a point just above the flare transition region, we obtain

$$\frac{3}{5}c_s^2 \ln\left(\frac{n_{\text{ch}}}{n_{\text{co}}}\right) + \frac{1}{2}(v_{\text{ch}}^2 - v_{\text{co}}^2) \geq -\int_{z_{\text{co}}}^{z_{\text{ch}}} \frac{\partial v}{\partial t} dz,$$

or

$$v_{\text{co}}^2 \leq \frac{6}{5} \ln\left(\frac{n_{\text{ch}}}{n_{\text{co}}}\right)c_s^2 + v_{\text{ch}}^2 + 2 \int_{z_{\text{co}}}^{z_{\text{ch}}} \frac{\partial v}{\partial t} dz, \quad (11)$$

where “co” refers to the ambient corona and “ch” to the evaporating chromospheric plasma.

We now argue that the expansion velocity into the residual

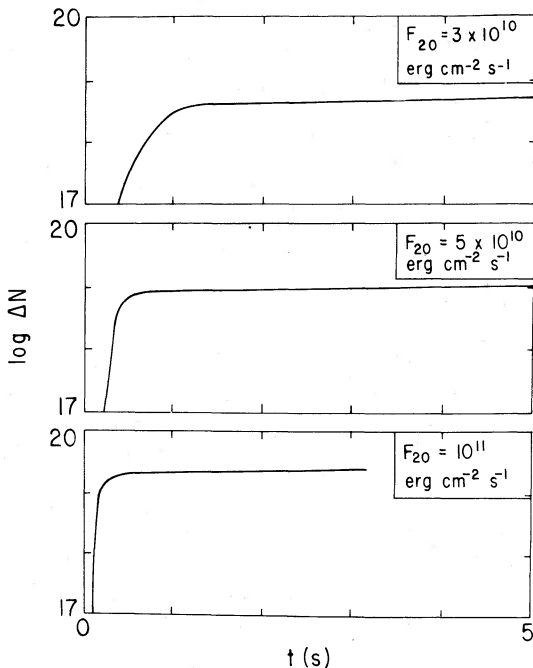


FIG. 2.—Column depth of evaporated material during flare heating. Values do not change noticeably after flare heating ceases at 5 s.

chromosphere, v_{ch} , is insignificant compared with v_{co} . Flow behind any shocks or disturbances driven by the expanding plasma travel with a speed that is roughly $v \approx (P_1/\rho_0)^{1/2}$, where P_1 is the pressure behind the shock, and ρ_0 is the ambient density ahead of it. Since the ambient density in the chromosphere is roughly 100 times larger than that in the corona, the upflow velocity v_{co} is roughly 10 times as large as the downflow velocity v_{ch} .

Next, we argue that during the initial period of expansion when the upward velocity is increasing, the time derivative $\partial v/\partial t$ is negative (since upward velocities are negative). These conditions allow us to write

$$v_{\text{co}}^2 \leq \frac{6}{5} \ln(n_{\text{ch}}/n_{\text{co}}) c_s^2 \equiv v_*^2, \quad (12)$$

where the right-hand side is clearly maximized if the preflare chromospheric density is used for n_{ch} .

Finally, we note that condition (8), the assumed monotonicity of T , can be relaxed to *nonmonotonic* temperature profiles, provided that

$$\int_{z_{\text{co}}}^{z_{\text{ch}}} n(dT/dz) dz \leq 0.$$

It can be shown that this inequality holds provided the sufficient (i.e., stricter) condition

$$T(z) \geq T_{\text{ch}} \quad (z_{\text{co}} < z < z_{\text{ch}}) \quad (13)$$

holds. Therefore, inequality (12) applies even if there are temperature peaks in the evaporating region, provided that condition (13) holds, and T_{max} (and c_s) is taken to correspond to the peak temperature, which may be greater than T_{co} .

The right-hand side of equation (12), which defines the upper limit v_* for upward expansion velocities, also shows that this limit is very insensitive to the density ratio. For a typical preflare loop, the density ratio $n_{\text{ch}}/n_{\text{co}}$ is about 10^2 , yielding $v_* = 2.35c_s$. Even if the density were taken 10 times larger, v_* would be increased by only 20%. We note also that T_{max} is usually close to the average coronal temperature, because of thermal conduction.

How useful an upper limit is v_* ? From our own simulations, and those of others, we find that the ratio v_{max}/v_* , where v_{max} is the maximum computed upward velocity in a given thick-target simulation, is between 0.6 and 1.0 for explosive evaporation, with v_{max}/v_* approaching unity for energy fluxes $\gtrsim 10^{11}$ ergs cm^{-2} s^{-1} . In the case of gentle evaporation, v_{max}/v_* is generally less than 0.3 and decreases with decreasing energy input flux. Thus explosive evaporation is significantly more effective at hydrodynamic acceleration than is gentle evaporation.

An additional consequence of the limit on upward velocities driven by chromospheric evaporation is that the strength of coronal hydrodynamic shocks driven by explosive evaporation is also limited. The evaporated material can be thought of as a piston which pushes coronal plasma ahead of it. The moving plasma is bounded in front by a shock which runs ahead of the piston. The speed of the front of evaporated material (and the plasma driven ahead of it), relative to the sound speed in the unshocked corona, defines the piston Mach number $M_p \equiv v_2/c_1$, where subscripts 1 and 2 refer to the material ahead of ($v_1 \equiv 0$) and behind the shock, respectively. Using conservation of mass, the coronal shock Mach number M can be related to the piston Mach number M_p by

$$M = M_p/(1 - n_1/n_2), \quad (14)$$

where we must regard n_1/n_2 as a function of M , the relation depending on the specific jump conditions appropriate to the shock. Since the upward expansion velocity, v_2 , is bounded by v_* , the piston Mach number is bounded by v_*/c_1 , which we denote M_p^{max} .

If the conductive flux into the top of the shock is sufficiently small that the ideal gas jump conditions apply, then the maximum shock Mach number, M_{max} , corresponding to M_p^{max} in equation (14), is given by

$$M_{\text{max}} = \frac{2}{3} M_p^{\text{max}} + [1 + (\frac{2}{3} M_p^{\text{max}})^2]^{1/2}. \quad (15)$$

If on the other hand, the conductive flux into the top of the shock is significant compared with the convective flow of thermal and kinetic energy through the shock front, and if the conductive temperature scale length is large compared with the shock thickness, we can take the temperatures on either side of the shock to be equal (Cargill and Priest 1982; Rosenau and Frankenthal 1976). In that case, equation (14) yields

$$M_{\text{max}} = \frac{M_p^{\text{max}}}{2} + \left[\left(\frac{1}{2} M_p^{\text{max}} \right)^2 + \frac{3}{5} \right]^{1/2}. \quad (16)$$

These limits may be useful in determining whether observed coronal shocks are consistent with being driven by chromospheric evaporation. For example, if the temperatures of the evaporated material, the shocked material, and the unshocked material are all approximately equal, then the Mach number limit given in equation (16), together with $v_* \approx 2.35c_s$, yields $M_{\text{max}} \approx 2.6$.

Finally, the upper limit to evaporation velocities provides us with a constraint on the range of validity for the gasbag model, discussed in the next section.

IV. THE GASBAG MODEL

In the previous section an upper limit to the evaporation velocity was derived, but its temporal evolution was not discussed. Dealing with this question on all relevant time scales of the flare is beyond the scope of the present paper. Here we concentrate on understanding the temporal behavior of evaporation velocities within times short compared with the hydrodynamic time scales of the flare loop as a whole. The dashed curve on the bottom panel of Figure 3 shows the evolution of the maximum upward velocity for the simulation with $F_{20} = 5 \times 10^{10}$ ergs cm^{-2} s^{-1} . Note that the velocity increases rapidly at first, and then after 3 or 4 s, more slowly. The maximum speed does not exceed 500 km s^{-1} during the 9 s of flare time simulated.

We can understand the temporal behavior of the velocity in this simulation, and in our other two explosive evaporation simulations, by the following idealized picture. Suppose that the initial density of the explosively evaporated region is uniform, and that at time $t = 0$, the temperature of this region is suddenly changed from its preflare chromospheric value to some coronal temperature. (Unfortunately, an analytic solution with a time-varying temperature cannot be obtained.) The subsequent evolution of the atmosphere is very similar to the behavior of a shock tube (Zel'dovich and Raizer 1967). The initial evolution is self-similar: from each end of the explosively evaporated region a rarefaction wave propagates into it at the sound speed, and the expanding gas then drives compression waves or shocks into the surrounding gas both above and below the region of enhanced pressure. Once the rarefaction waves meet at the center of the explosively evaporated region,

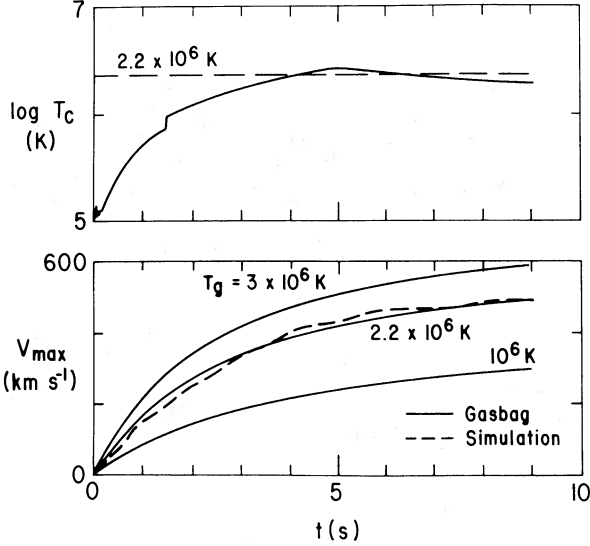


FIG. 3.—The solid curve in the upper panel is the average temperature in the explosively heated region for the $F_{20} = 5 \times 10^{10}$ ergs cm $^{-2}$ s $^{-1}$ simulation, while the dashed curve shows the most compatible temperature used in the gasbag model. The dashed curve in the bottom panel is the maximum upward velocity seen in the simulation, while the solid curves represent predicted maximum velocities for three differing gasbag temperatures, T_g .

the evolution is no longer self-similar, but the basic physical picture of the expansion remains valid.

In this paper, we are primarily concerned with the upflowing material. We pointed out in § III that the expansion velocity into the residual chromosphere is small compared with upward velocities. Therefore, an appropriate description of the expansion may be obtained by taking the velocity at the base of the explosively heated region to be zero. Since the explosively heated region is initially quite thin (a few hundred kilometers), the rarefaction waves propagate through it in $\lesssim 1$ s. Therefore, we neglect the internal dynamics of the heated region and assume that the density in the evaporating region stays uniform, while decreasing as the explosively heated material expands. These two assumptions, together with the continuity equation (20), imply that the velocity in the explosively heated plasma varies linearly in space and may be written

$$v(N, t) = v_1(t)N/\Delta N, \quad (17)$$

where $v_1(t)$ is the velocity at the top of this “gasbag,” ΔN is the column thickness of the gasbag, and N is measured upward from the base of the gasbag. We will consider only the expansion of the gasbag itself, neglecting the inertia of shocked coronal material ahead of the gasbag. Therefore, our description breaks down once a mass of plasma comparable to the mass of the explosively heated region (gasbag) has been swept up. Inserting equation (17) into the momentum equation

$$m\partial v/\partial t = -\partial P/\partial N, \quad (18)$$

and integrating from the base of the gasbag, where the density and pressure are n_g and P_g , to the undisturbed corona ahead of the shock (density and pressure n_{co} and P_{co}) and neglecting the influence of shocked material ahead of the gasbag, we obtain

$$\frac{1}{2}m\Delta N(\partial v_1/\partial t) = P_g - P_{co} = P_g(1 - n_{co}/n_g), \quad (19)$$

where we have used the assumption $T_g = T_{co}$, where T_g is the constant temperature inside the gasbag, and T_{co} is the ambient coronal temperature.

In understanding the nature of the approximations we have made, it is useful to note that equation (19) is the equation of motion of a piston of mass $\frac{1}{2}m\Delta N$, that is, half the total mass of the gasbag, driven by the expansion of a massless gas of pressure P_g against an external pressure P_{co} . The inertial mass of the equivalent piston is half the mass of the gasbag because, with a linear velocity profile, the average velocity of the gas is just half of v_1 .

Turning now to the continuity equation, which can be written

$$\partial(1/n_g)/\partial t = \partial v/\partial N, \quad (20)$$

differentiating with respect to time, and substituting for $\partial v/\partial t$ using equations (17) and (19), we find

$$\partial^2/\partial t^2(1/n_g) = P_g(1 - n_{co}/n_g)/(\frac{1}{2}m\Delta N^2). \quad (21)$$

This equation may be simplified by using the length $L(t)$ of the gasbag as the dependent variable, where $n_g L \equiv n_{g0} L_0$ by conservation of mass, where n_{g0} and L_0 are the initial gasbag density and length, respectively. Defining the quantity

$$c_T = [P_g/(mn_g)]^{1/2} = (2kT_g/m)^{1/2} \quad (22)$$

as the coronal isothermal sound speed, equation (21) may be written

$$d^2 L/dt^2 = 2(c_T^2/L)[1 - (n_{co}L)/(n_{g0}L_0)] \quad (23)$$

and integrated to give

$$\frac{1}{2}(dL/dt)^2 = 2c_T^2[\ln(L/L_0) - (n_{co}/n_{g0})(L/L_0 - 1)], \quad (24)$$

which defines the velocity of the front of the gasbag $v_1 \equiv dL/dt$ as a function of its length. Since this equation breaks down once substantial expansion has taken place ($L/L_0 \gg 1$), as a result of our neglect of the coronal plasma accelerated ahead of the gasbag, we can examine only the initial evolution. Therefore, we may neglect also the coronal back pressure, represented by the term n_{co}/n_{g0} in equation (24), and integrate again to obtain the equation describing the expansion of the gasbag, restrained only by its inertia:

$$\int_0^{\eta(t)} \eta^{-1/2} e^\eta d\eta = 2c_T t/L_0, \quad (25)$$

where $\eta(t) \equiv \ln(L/L_0)$. From equation (24) we see that (neglecting the term n_{co}/n_{g0})

$$v_1(t) = 2c_T \eta^{1/2}(t). \quad (26)$$

The quantities $\eta(t)$, $e^{-\eta(t)}$, and $\eta^{1/2}(t)$ are shown in Figure 4.

We can obtain a limit on the duration of validity of the gasbag model by requiring that the maximum velocity $v_1(t)$ be less than the upper limit to evaporation velocities derived in the previous section. Requiring $v_1 \leq v_*$ yields

$$\eta(t) \leq \frac{1}{2} \ln(n_{ch}/n_{co}), \quad (27)$$

or $\eta(t) \leq 2.3$, for $n_{ch}/n_{co} \approx 100$. From Figure 4, this implies that the gasbag solution is approximately valid for times $t \leq 4L_0/c_T$. At this time the gasbag has expanded to $L/L_0 = e^\eta \approx 10$ times its original size. One useful property of the gasbag model is that it shows that the expansion time scale is proportional to L_0/c_T . Defining the characteristic expansion time τ to be the time for an e -fold expansion (i.e., $\eta = 1$), we find from Figure 4 that $\tau \approx 1.5L_0/c_T$.

We now wish to compare the results of the gasbag model with those of our simulations. Taking the case of $F_{20} =$

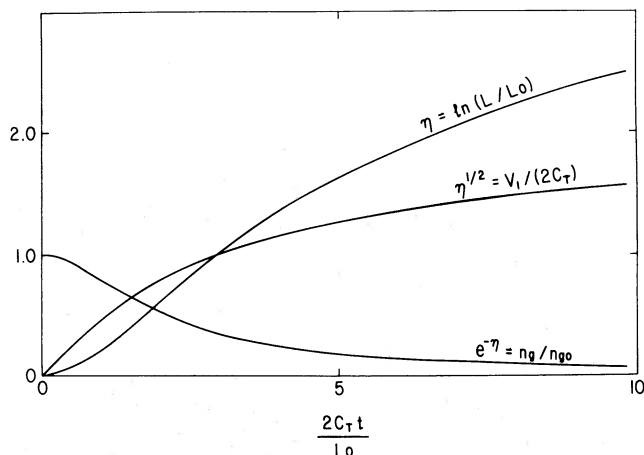


FIG. 4.—The gasbag model solution curves. Here, L_0 is the initial length of the gasbag, and c_T is the isothermal sound speed corresponding to the gasbag temperature T_g .

$5 \times 10^{10} \text{ ergs cm}^{-2} \text{ s}^{-1}$ as an example, we find from the simulation that the column depth of explosively evaporated plasma is $\Delta N = 1.1 \times 10^{19} \text{ cm}^{-2}$, and that the average preflare density in this region is $n_{g0} \approx 4.2 \times 10^{11} \text{ cm}^{-3}$, so that $L_0 \approx 260 \text{ km}$. Since the heating time scale is *not* in fact short compared with the expansion time scale, as our idealized analysis assumes (see top panel of Fig. 3), the appropriate choice of a gasbag temperature is not obvious. Hence we let T_g be a free parameter and choose the value which best fits the temporal behavior of the maximum upward velocity seen in the simulation. This value can then be compared with the actual temperature of the evaporated material in the simulation. For the case of $F_{20} = 5 \times 10^{10} \text{ ergs cm}^{-2} \text{ s}^{-1}$, the optimum value of T_g is $2.2 \times 10^6 \text{ K}$. The comparison of the gasbag solution to the simulation result is quite striking (see Fig. 3). Not only is the character of the time dependence very similar, but the actual magnitudes of the velocities are very close. Moreover, the value of T_g is a good approximation to the average temperature if the early part of the expansion ($t \leq 3 \text{ s}$) is excluded. Equation (23) remains valid if a time-dependent temperature is used [$c_T^2 \propto T_g(t)$], but it cannot then be integrated analytically.

As an additional test, we compare the gasbag model with the maximum velocity in the flare simulation of MacNeice *et al.* (1984). They assumed a flux of nonthermal electrons of $10^{11} \text{ ergs cm}^{-2} \text{ s}^{-1}$, with a low-energy cutoff of 10 keV, lasting for 10 s, as compared with our 5 s. The longer heating time and

lower cutoff energy at this high energy flux produced much hotter explosively evaporated plasma than did any of our calculations. The initial spatial extent of their explosively evaporated region (L_0) encompasses the top 750 km of their preflare chromosphere, and the temperature of the evaporated plasma saturates at $T_g \approx 14 \times 10^6 \text{ K}$ after 2 or 3 s. With these parameters, the gasbag model matches their numerical simulation fairly well from 2 to 8 s (see Fig. 5). It is clear from their velocity plots that free expansion of the explosively evaporated plasma ceases at about 9 s, when the gasbag hits the top of their loop.

The gasbag model is useful conceptually in giving a simple picture of the early stage of explosive evaporation and, moreover, has succeeded in reproducing the thermal velocity behavior of two independent numerical simulations of explosive evaporation. We caution, however, that the model is not entirely self-consistent and therefore is not necessarily generally applicable.

V. EVOLUTION OF THE FLARE TRANSITION REGION

One of the most distinctive features of explosive evaporation is that the entire upper chromosphere is raised from chromospheric to coronal temperatures nearly simultaneously. Thus for a short period of time a very large amount of plasma exists at temperatures around 10^5 K . At this time, emissions normally seen in the very narrow transition region should be very much enhanced, provided that the ionization and recombination times are short enough. Once the temperature of the evaporated plasma has been raised above 10^5 K , however, one would expect transition region emissions to decrease, finally reaching a quasi-steady value depending on the structure of the newly formed transition region. This behavior can be clearly seen in Figure 6, where we plotted the differential emission measure $\xi(T)$ [$\xi(T) \equiv n_e^2 |d \ln T / dz|^{-1}$] at $T = 10^5 \text{ K}$ for $F_{20} = 5 \times 10^{10} \text{ ergs cm}^{-2} \text{ s}^{-1}$, a case of explosive evaporation. Note that $\xi(10^5 \text{ K})$ jumps from a preflare value of less than 10^{27} cm^{-5} to over 10^{31} cm^{-5} in about 0.1 s! The differential emission measure then falls slightly, until at 1.4 s, a sudden spike occurs, followed by a drop to about 10^{29} cm^{-5} . The spike coincides with the formation of a chromospheric condensation, which will be discussed in Paper VII (Fisher, Canfield, and McClymont 1985b). The value of $\xi(10^5 \text{ K})$ then stays fairly constant until the flare heating is turned off at 5 s. After that, it falls in a roughly exponential manner with time.

The evolution of differential emission measure seen here differs from that found by Emslie and Nagai (1985). Their $\xi(10^5 \text{ K})$ for thick-target heating increases monotonically in

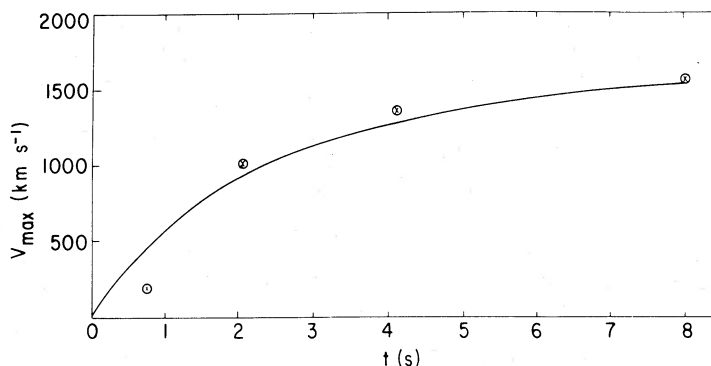


FIG. 5.—Gasbag model with $L_0 = 750 \text{ km}$ and $T_g = 14 \times 10^6 \text{ K}$ (solid line), compared with maximum upflow velocities from MacNeice *et al.* (1984) (points)

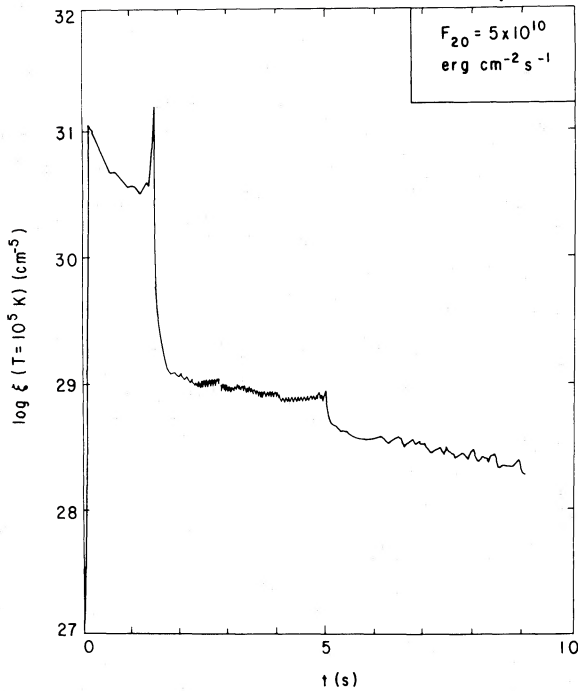


FIG. 6.—Differential emission measure $\xi(T) \equiv n_e^2 |d \ln T/dz|^{-1}$ at $T = 10^5$ K as a function of time for $F_{20} = 5 \times 10^{10}$ ergs $\text{cm}^{-2} \text{s}^{-1}$

time, for both parallel and isotropic electron injection. We believe the discrepancy is due to differences in the temporal variation of the flare heating input between our simulation and theirs. The flare heating input rate in our calculation remains constant during the 5 s burst, while Emslie and Naglai (1985) assume the flare heating rate ramps up with a 30 s time scale. Thus, the flare transition region column depth N_1 in equation (2) moves steadily downward in their simulation as the heating rate increases. Since their 30 s heating rise time is far longer than the explosive evaporation temperature rise time, they see no short pulse in $\xi(10^5 \text{ K})$ corresponding to the heating *en masse* of the entire upper chromosphere.

An issue of importance in the evolution of the transition region and corona during flares is the extent to which the conductive flux approaches the free-streaming limit. Our treatment of thermal conduction described in § V of Paper V ensures that the conductive flux is bounded by the free-streaming limit. We have plotted in Figure 7 the ratio F_{cl}/F_{sat} as a function of temperature for selected times during the $F_{20} = 5 \times 10^{10}$ ergs $\text{cm}^{-2} \text{s}^{-1}$ simulation, where F_{cl} is the Spitzer-Härm (classical) conductive flux, and F_{sat} is our adopted free-streaming (saturated) limit (eq. [17] of Paper V). Note that

1. saturation ($F_{cl} \approx F_{sat}$) does occur at coronal temperatures ($T \gtrsim 10^{6.5} \text{ K}$), but

2. the conductive flux is at least an order of magnitude below the saturation limit for all temperatures $\lesssim 10^6 \text{ K}$, and a factor of 30 or more below the limit for $T \lesssim 10^5 \text{ K}$.

Since thermal conduction in the lower transition region can thus be treated classically in these simulations, it is straightforward to relate the conductive flux at 10^5 K , F_5 , to the differential emission measure $\xi(10^5 \text{ K})$. The behavior of the conductive flux in the lower transition region can be used to investigate the

processes most important in determining its structure. Expressing the conductive flux in terms of $\xi(T)$, we have

$$F_c \equiv -\kappa_0 T^{5/2} (dT/dz) = n^2 \kappa_0 T^{7/2} / \xi(T). \quad (28)$$

The simulations show that F_5 is inversely correlated with $\xi(10^5 \text{ K})$ (Fig. 6), as suggested by equation (28) if the variation of $n(T = 10^5 \text{ K})$ is neglected. The flux F_5 initially decreases from its preflare value, reflecting the shallow temperature gradient corresponding to the heating *en masse* of the explosively evaporated chromosphere. A sharp dip in F_5 then occurs, coinciding with the formation of the chromospheric condensation, mentioned above, after which F_5 rises again, reflecting the temperature gradient of the newly formed flare transition region.

The quasi-steady value of F_5 that is reached after the initial adjustment period can be understood very simply in terms of the pressure in the flare transition region. First of all, after the initial adjustment period, the transition region is very thin, so the pressure must be nearly uniform across it. Using this result, and the assumption that the structure of the lower transition region ($T \leq 10^5 \text{ K}$) is determined by a balance between conductive heating and radiative losses (verified by the simulations), we can write

$$-dF_c/dz = n^2 f(T) \equiv [P/(2kT)]^2 f(T), \quad (29)$$

where F_c is the conductive flux, defined in equation (28), with $\kappa_0 \approx 10^{-6}$ (cgs units). Combining equations (28) and (29) and integrating, we find

$$F_c(T) = (2\kappa_0)^{1/2} (P/2k) \lambda(T)^{1/2}, \quad (30)$$

where

$$\lambda(T) = \int_{T'}^T f(t) t^{1/2} dt. \quad (31)$$

We have assumed that at some chromospheric temperature T'

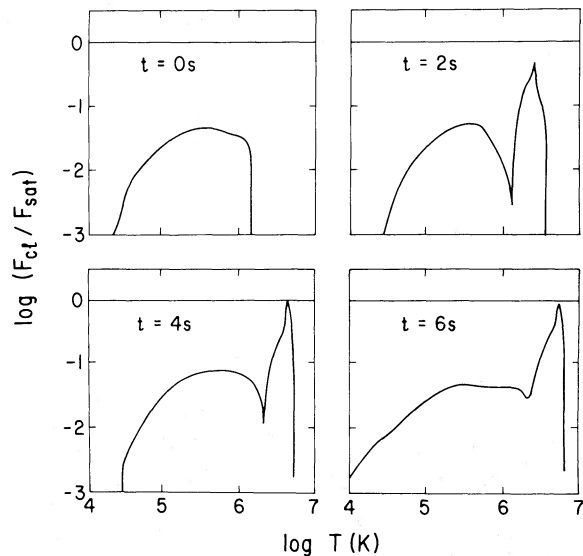


FIG. 7.—The ratio of the classical heat flux to the saturated flux limit as a function of temperature at four selected simulation times (for $F_{20} = 5 \times 10^{10}$ ergs $\text{cm}^{-2} \text{s}^{-1}$). Low values of the ratio correspond to regions where the Spitzer conductivity is valid. Conduction remains classical in the transition region ($T \lesssim 10^6 \text{ K}$) but is saturated in the corona ($T \gtrsim 10^{6.5} \text{ K}$) at $t = 4 \text{ s}$.

the conductive flux is negligibly small compared with that at T . Approximating our optically thin radiative loss function $f(T)$ by (McClymont and Canfield 1983)

$$f(T) = 7 \times 10^{-37} T^3 \text{ ergs cm}^3 \text{ s}^{-1} \quad (32)$$

for temperatures less than 10^5 K, and evaluating equation (30) at $T = 10^5$ K, we find

$$F_5 = 3.42 \times 10^5 P \text{ ergs cm}^{-2} \text{ s}^{-1}, \quad (33)$$

where P is the transition region pressure (dyn cm^{-2}). This argument, similar to that of Moore and Fung (1972) and Ricchiazzi (1982), shows that the conductive flux at 10^5 K is just proportional to the pressure. We note that the quasi-steady value of F_5 obtained in equation (33) can be used to derive a similar expression for the quasi-steady behavior of $\xi(10^5 \text{ K})$ through equation (28); namely,

$$\xi(10^5 \text{ K}) = 1.21 \times 10^{27} P \text{ cm}^{-5}. \quad (34)$$

The quasi-steady value of F_5 that is reached during flare heating is plotted as a function of pressure in Figure 8, for several simulations. Note that F_5 is indeed closely proportional to the pressure as equation (33) predicts. Thus, even when evaporation is driven conductively, there is still a close balance between conductive heating and radiative losses in the lower transition region (see Craig and McClymont 1984).

We now compare the results of our calculations with *SMM* observations of O v emission in several events presented by Poland *et al.* (1984). We have not computed O v fluxes from our models, but we anticipate that the behavior of $\xi(10^5 \text{ K})$ in our calculations should be quite similar to that of computed fluxes of O v line emission, which is formed at temperatures around 2.5×10^5 K.

The differences between the observed enhancement of O v emission and $\xi(10^5 \text{ K})$ for the $5 \times 10^{10} \text{ ergs cm}^{-2} \text{ s}^{-1}$ simulation (Fig. 6) are profound. Whereas the observed O v emission increases by modest amounts (factors of 2–3 according to Poland *et al.*) during flares, calculated values of $\xi(10^5 \text{ K})$ increase by four orders of magnitude during explosive evaporation. Moreover, even during the later quasi-steady phase,

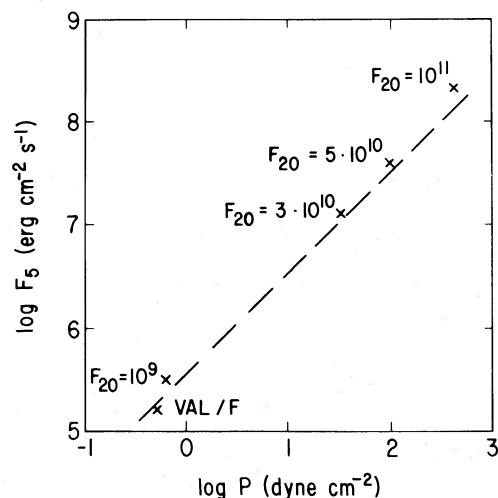


FIG. 8.—The conductive flux in the quasi-steady flare transition region at $T = 10^5$ K, F_5 , as a function of transition region pressure, for several simulations. The dashed line represents the quasi-steady relationship $F_5 = 3.42 \times 10^5 P$.

there is still 100 times more emission at 10^5 K than there is before the flare. If we are to interpret the observations of Poland *et al.* (1984) in terms of nonthermal electron heating in loops, then the following conclusions can be drawn:

1. None of the flares observed by Poland *et al.* exhibit explosive evaporation, since the observed emission does not have the characteristic large emission spike seen (for example) in Figure 6. Therefore, we may conclude that the flux of energetic electrons is low, or the rise is long (cf. Emslie and Nagai 1985), and that the hydrodynamic evolution can be described in terms of gentle evaporation; furthermore, the state of the transition region can be described in terms of the quasi-steady equations (29)–(34).

2. The O v flux, to the extent that it is described by $\xi(10^5 \text{ K})$, should be proportional to the pressure in its region of formation (see eq. [34]). Since the flux increases by only a factor of 2–3, we conclude that the transition region pressure increases by only this amount.

3. Flares for which the O v emission shows little or no temporal hysteresis when plotted against the hard X-ray flux imply that very little chromospheric evaporation takes place. The final transition region pressure is the same as its preflare value, as in our $F_{20} = 10^9 \text{ ergs cm}^{-2} \text{ s}^{-1}$ simulation.

4. The large range of O v fluxes in different flares, which have comparable hard X-ray fluxes (Fig. 3 of Poland *et al.*), can be easily explained if the flares take place in loops of different initial pressures (as suggested by Poland *et al.*).

5. Flares which do show temporal hysteresis in O v versus hard X-rays (i.e., curve 2 of Fig. 3 in Poland *et al.* 1984) are consistent with a moderate degree of chromospheric evaporation, giving rise to the inferred increase in pressure during the flare.

The above conclusions could be tested for consistency if simultaneous soft X-ray observations of the flares by Poland *et al.* (1984) were available. We would expect, for example, to find no increase in soft X-ray emission after the impulsive bursts which do not show hysteresis in O v versus hard X-rays, while a coronal density enhancement should accompany a long-term increase in the transition region pressure (and flux of O v).

Finally, we can use the simulations to examine the hydrodynamic velocity in the flare transition region. In our simulations of *explosive* evaporation, we find that the velocity at transition region temperatures is downward (see Fig. 3 of Paper VII, for example). For the case $F_{20} = 10^{11} \text{ ergs cm}^{-2} \text{ s}^{-1}$, for instance, the downward velocity at 10^5 K is near 40 km s^{-1} . However, in the case of *gentle* evaporation—for example, $F_{20} = 10^9 \text{ ergs cm}^{-2} \text{ s}^{-1}$ —the velocity is upward, at about 20 km s^{-1} . The reason for this fundamental difference in sign of the velocity is discussed in Paper VII.

It is interesting to compare these results with observations of transition region velocities in the 1980 April 8 flare made by Cheng *et al.* (1982). At the onset of the flare, velocities inferred from transition region lines were in fact downward, at about 40 km s^{-1} . In addition, the densities inferred from these observations showed a definite jump to about 10^{12} cm^{-3} , an order of magnitude increase. Both of these features suggest that this flare undergoes explosive, as opposed to gentle, evaporation, if we can interpret the flare as being heated by nonthermal electrons.

VI. CONCLUSIONS

We have demonstrated that there is an energy flux threshold of importance to the hydrodynamic response of the plasma in

flares heated impulsively by nonthermal electrons. Above this threshold, evaporation is explosive and very effective at driving coronal mass motion at high velocities. Below the threshold, evaporation is gentle, and the driven velocities are relatively low. The direction of the transition region velocity also changes at the threshold, being upward for low heating fluxes and downward for high fluxes. The threshold can be determined by equating the thick-target heating rate per particle with the peak radiative loss rate (at $T \approx 10^5$ K) at the chromosphere just below the preflare transition region.

We show that there is an upper limit to coronal velocities driven by chromospheric evaporation. This upper limit is sensitive primarily to the temperature of the evaporated material, which allows a test of consistency of observations with the chromospheric evaporation model. We show elsewhere (Fisher, Canfield, and McClymont 1984) that upward velocities observed by *SMM* (Antonucci and Dennis 1983) are less than this upper limit and hence consistent with chromospheric evaporation. Nonhydrodynamic (e.g., magnetic) forces need not be invoked to explain the observed velocities. We find that the maximum computed velocities in our numerical flare simulations approach this upper limit for explosive evaporation driven by large electron fluxes, while for gentle evaporation, maximum velocities are less than $\sim 30\%$ of the upper limit.

We derive a simple model for explosive evaporation based on the isothermal expansion of a mass of hot plasma. This gasbag model is valid for times short compared with hydrody-

namic time scales in the coronal loop. The time scale for expansion in this explosive phase is given by $\tau \approx 1.5L_0/c_T$, where L_0 is the initial extent of the explosively heated region, and c_T is the isothermal sound speed in the heated region. The temporal behavior of the upward velocity given by the gasbag model agrees well with that of our numerical simulations when the gasbag temperature is close to the temperature of the evaporated material in our simulations.

Finally, we discuss the evolution of the lower transition region during chromospheric evaporation. In the case of explosive evaporation, a large amount of plasma is heated from chromospheric to coronal temperatures *en masse*, resulting in a very large, short-lived, emission measure at transition region temperatures. This may result in a short, powerful EUV flash. Once a quasi-steady equilibrium is established in the flare transition region, the conductive flux at 10^5 K is just proportional to the transition region pressure. When we compare the results of our numerical simulations with observed transition region line emission, we find that the observations of Poland *et al.* (1984) are most consistent with heating by *small* fluxes of energetic electrons, and therefore gentle evaporation, while the observation of Cheng *et al.* (1982) may be an example of explosive evaporation caused by a *large* flux.

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